Remote Sensing

Introduction

In this study, we will test various remote sensing processing methods that will be used to train the model for detecting the 2026 Gironde wildfire using satellite data and remote sensing processing techniques (indices, color composites), statistical methods (PCA, MNF) to generate data for training a machine learning model of classification to detect and classify the trace of the wildfire and models of Deep Learning (Markov Chains) and Cellular Automata (CA-Markov) to modelling the future regeneration of the vegetation in the zone .

We will examine the Gironde fire, specifically the portion that affected Le Porge and Lège-Cap-Ferret, extending to the outskirts of the Bordeaux metropolitan area which began on July 21 and ended on August 8.

II) Issues

What remote sensing data and processing methods can be used to train a machine learning model to detect forest fires , estimate their intensity and simulate the regeneration of the vegetation and agricultural parcels ?

III) Types of data

The types of data are as follows :

- **Satellite images** : Landsat 8-9 OLI-TIRS before the wildfire (June 22 , 2026) and after the wildfire (August 8 , 2026)
- **Vector data** : layer of the trace of the wildfire

IV) Methodology

V) Elements of Methodology

In the workflow of methodology , we realised multiple tests for each processing to determine the best approach for creating data for the Machine Learning Model and after the Deep Learning model of prediction :

Calculation of Indices and ratio :

calculation of the NDVI (Normalized Difference Vegetation Index) / CTVI (Corrected Transform Vegetation Index)

Color Composite

Blue Band , Near-Infrared Band , Thermal Infrared band / NDVI , Near-Infrared Band , Thermal Infrared Band / CTVI , Near-Infrared Band , Thermal Infrared Band

PCA

with the Blue Band of the spectrum of the visible , Red band of the spectrum of the visible , Near-Infrared Band and Thermal infrared 1 band

PCA

with different analysis types (T-Mode , S-Mode , T and S Mode , Inverse T-Mode , Inverse S-Mode) and different matrix type (Covariance Matrix , Correlation Matrix , Cross-Product , Normalized Cross-Product) and we choose the four components of the T and S mode and the Normalized Cross Product

MNF

we test the MNF with the different noise estimation methods (Adjacent Lower Cell , Bottom right diagonal cell , Adjacent Right Cell , Upper right diagonal cell) with the component 2 and 4 of the PCA

Filters of Convolutions :

Kernel Directional Filter / Laplacian Enhancement Filter / High Pass Filter

Machine Learning algorithms of Unsupervised classification :

K-Means (with different cluster centroid Initialization : Random Seed / Diagonal Axis /Random Partition) / Isoclust with 10 or 15 and 500 iterations for each training of the models

Prediction and Modelisation with Deep Learning and Cellular Automata :

models like Markov Chains with four iterations for the four the seasons and 20 Iterations for 20 years and and CA-Markov Chains Model (Composed with the K-Means after the fire , the Markov Transition areas image and the Markov Probability folder) with cellular automata iterations to simulate the fours seasons of 20 years (80 cellular automata iterations) ,30 (120 cellular automata iterations) after and a standard 5X5 Contiguity Filter to Modelling the reconstruction of the vegetation 20 years and 30 years (with an analysis of the four seasons) after the fire

V) Analysis

- The NDVI , CTVI and color composites allow for the detection of the trace of the Wildfire of Gironde with the processing before the wildfire (June 22 , 2026) and the condition of the vegetation with a decrease of the NDVI Values after the wildfire in the zone (decreasing from 0.93 before the wildfire to 0.17 after the wildfire)
- The difference in NDVI between before and after the wildfire permits to quantify the loss of vegetation.
- The NDVI permits us to see the trace of the wildfire more clearly than the CTVI, which has less contrast between the burned zone and the vegetation.e
- The PCA with the S and the T-Mode allow us to visualize more precisely the trace of the wildfire especially with the component 2 and 4 which allows to visualize the trace and the other do not permit to observe the Wildfire (Component 1 (A1) and Component 3(A2))
- The MNF Bottom Right diagonal cell with the Component 2 which allow for detect the intensity of the wildfire and the component 4 which allow to detect the parcels , the component 1 (A3) because she is less precise and blurred and component 3 (A4) are not really interesting for the detection of the most affected zones data in model
- The Directional convolutional filters , allow for distinguishing more precisely than

the NDVI MNF and PCA the plots which are touched by the Wildfire , and more precisely the component 2 of the Kernel NSEW(A5) which permit to detect the zones more or less affected by the wildfire and the component 4 which makes it possible to detect the parcels . The Kernel NSEW 3X3 (Band 1 (A6) and Component 3 (A7)) , 5X5(A8) (four bands (A9)(A10)(A11)(A12)) , Laplacian Filter in 3X3 (A13)(A14)(A15)(A16) , 5X5 (A17)(A18(A19)(A20) and 7X7(A21)(A22)(A23)(A24) (four bands) and High-Pass Filter 3X3(A25)(A26)(A27(A28) , 5X5(A29)(A30)(A31)(A32) , 7X7(A33)(A34)(A35)(A36) are not really precise to detect the trace of the wildfire

- The different Machine Learning models trained on Cluster data , Isoclust , the Normalized Cross Product (four bands) ; MNF (Component 2 and 4) , Kernel Convolutional filter (NSEW) 5X5 (Component 2 and 4) both unsupervised (K-Means , Isoclust) allows us to classify the zones which are more touched by the wildfire and more specifically the K-Means which permit to detect all the parcels and the trace (with the cluster 1 in white) and the zones which are affected and parts of parcels which are more affected than others (with the cluster 2 in Green) .
- The folder of probability of changing of the Markov (created with the K-Means befor and after the fire) permit to visualize the predictive scenario of an return of the vegetation in the zone of the fire (class 6) because we have an return of 40% of the class 4 who correspond to dry vegetation in the zone of the fire and also an important evolution (9,63¨%) of the class 9 in the class of the fire who correspond to a development of an very , the return of the the class 8 (8,82%) correspond to the regeneration of a wet vegetation and a little return of the agricultural parcels in the zone of the fire (2,13%) , the image of the class 6 permit to visualize a first step of the return of the vegetation in the fire with a diminution of the trace in the two simulations
- The Modelling with models of Deep Learning like Markov Chains and CA-Markov permit to modelling the redevelopment of the vegetation the zone in the zone 20 years after the fire permit to saw a partial regeneration in the zone of the wildfire with a part of very dry vegetation (represent with the class 4 l) , a little regeneration of the the agricultural parcels (class 11) , thirty years after we saw an more important regeneration of the vegetation and the agricultural parcels (class 11) and a development of wet vegetation (Class 8 and 9) in the zones of dry vegetation (class 5) .

VI) Interpretation and hypothesis

Inside the trace, the zones not burned keep a high NDVI, and we can see circles of irrigation in the component 2 of the PCA. So they are probably agricultural plots. They were possibly also protected by the firefighters.

The results of all processings steps permit the visualization of the zones most affected by the Wildfire with the Minimum Noise Fraction first , the Convolutional filters and the Machine Learning classification processings (K-Means) which allow for classify more simply the zones which are most affected by the wildfire, they may correspond to agricultural plots or unburned vegetation.

This also enables the visualization of unburned zones , which likely correspond to urban areas exhibiting higher resilience to the disaster (This was more precisely the case with the central right zone which is not burned and which is predominantly composed of urban zones).

The CA-Markov model provides a possible scenario of vegetation changes 20 years after the

wildfire. The results suggest that different land cover classes could evolve differently over time.

VII) Limits of the work

The limitations of this work are :

- The spatial resolution of Landsat 8-9 *OLI TIRS* which is less precise than *Sentinel-2 , Sentinel-1 SAR* , *Pléiades Neo* or *HRI* does not allow for the precise detection of all wildfire details and its intensity.Sentinel-1 SAR would allow to see the zones through the smoke.
- The study uses only one satellite image before the wildfire and one after the event. This limits the analysis of vegetation changes over time
- The use of Digital Elevation Model (DEM) would have enabled us to analyze the slope and roughness , which help visualize zones more or less affected by the wildfire.
- The CA-Markov model is based on a limited number of satellite images and does not fully represent the long-term dynamics of vegetation recovery.

VIII) Conclusion

First this workflow combining remote sensing , statistical analysis and machine learning permits to detect zones using NDVI and thermal-band color composites to create a dataset and a model permit to create different processings which use different methods , like remote sensing with color composite , indices and ratio , statistical analysis (with the MNF and the PCA).
Convolutional filters to create a model of detection of a wildfire and to detect the zones who are more and more less affected by the wildfire.
The CA-Markov model provides a possible scenario of future vegetation changes after the wildfire. However, additional satellite images and validation data would be necessary to improve the reliability of the results and better understand long-term vegetation recovery.

IV) Methodology
V) Elements of methodology
VI) Results (Before the fire)
VII) Results (After the fire)
VIII) Results (Before and After the fire) : Unsupervised Classification with Machine Learning models
IX)